Overview

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Study / Model	Year	Clinical Task	Input Modality	Architecture	Key Outcome / Accuracy
Ghafoori & Shalbaf	2025	pMCI vs. sMCI	**rs-fMRI** + Clinical	**3D-CNN + LSTM**	**92.47% Acc.** High performance achieved by fusing spatial (CNN) and temporal (LSTM) features.
Sun et al.	2025	pMCI vs. sMCI vs. rMCI	**rs-fMRI**	**LSTM-lsGC**(Effective Connectivity)	**90.86% Acc.** (rMCI vs pMCI). Introduces causal connectivity & reversion as key concepts.
Zhang & Jiang (STGC-GCAM)	2025	pMCI vs. sMCI	**rs-fMRI**	**Spatiotemporal GCN** + Grad-CAM	**0.85 Accuracy.** Validated on large multi-site data (N=2272); highly interpretable via heatmaps.
DAFN	2024	pMCI vs. sMCI	MRI + Cognitive	**3D-CNN + Transformer**	**81.34% Acc.** Uses cognitive scores to *guide* the visual attention of the CNN.

1.2 The Evolution of Computational Approaches

The trajectory of machine learning in this domain has been steep.

- **Generation 1 (2010-2016):** Traditional Machine Learning. Researchers extracted Regions of Interest (ROIs), computed static Pearson correlation matrices, and fed the upper triangle of these matrices into Support Vector Machines (SVMs) or Random Forests. These methods ignored the temporal dynamics of the brain and treated connectivity as a static fingerprint.
- **Generation 2 (2016-2020):** 2D and 3D Convolutional Neural Networks (CNNs). Studies treated fMRI connectivity matrices as images or utilized 3D CNNs on voxel-wise data. While powerful, CNNs struggle with the non-Euclidean nature of brain data; two spatially distant regions (e.g., left and right parietal lobes) might be functionally connected, but a standard convolution kernel cannot easily capture this relationship.
- **Generation 3 (2021-2023):** Recurrent Neural Networks (RNNs) and basic Graph Neural Networks (GNNs). These models began to address temporal dynamics (via LSTMs) and topology (via GNNs).
- **Generation 4 (2024-Present):** The current SOTA. The field is now dominated by **Dynamic Graph Neural Networks (dGNNs)**, **Transformers**, and **Generative Diffusion Models**. These architectures are designed to handle 4D data (3D space + Time) without collapsing the temporal dimension, effectively modeling the brain's "metastates".

The 2024-2025 literature emphasizes that the brain is not a static graph but a dynamic system that fluctuates between varying states of integration and segregation. Models that capture this *chronnectome* (time-varying connectome) consistently outperform static baselines.

Dynamic Graph Neural Networks (dGNNs): The 2024 Paradigm Shift

The most significant cluster of high-impact papers in the review period focuses on Dynamic Graph Neural Networks. These models depart from the assumption of Static Functional Connectivity (sFC), which averages neural activity over the entire scan duration (typically 5-10 minutes). Evidence suggests that critical biomarkers of AD—such as the "dwelling time" in specific network states—are lost in this averaging process.